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ON EINSTEIN-MINKOWSKI SPACE-TIME

TWO recent articles—by C. W. Rietdijk * and Hilary Putnam †—draw philosophical conclusions of a rather strong character from the structure attributed to space-time by the special theory of relativity. The latter is by now so well understood from a mathematical point of view, and has been so often discussed, that one might have thought further elementary exposition redundant. However, by a curious coincidence, the two articles cited draw their conclusions by means of the same basic argument: a very simple, and in a certain geometrical sense a correct, one; but an argument in both cases seriously misapplied (and in essentially the same way), so that the asserted conclusions do not follow. This has led me to believe that one more account of the rudiments of Einstein-Minkowski space-time theory may not after all be one too many. Such an account is contained in the next section. The subsequent sections deal, first with the common basic fallacy of the two articles, and then with certain further issues arising from Putnam's paper.

I

The crucial lesson of Minkowski's contribution to relativity (a lesson that has in fact become a commonplace in contemporary mathematics) is this: the best understanding of any geometrical structure—such, for example, as the space-time of Einstein's special theory—is to be obtained by studying the geometrical properties and relations *intrinsic* to that structure, in preference to the equations or other formulas that express those properties by reference to more or less arbitrary auxiliary constructs. In so far as the latter have real importance, they will appear in the "intrinsic" theory as expressing *relations,* in-

* "A Rigorous Proof of Determinism Derived from the Special Theory of Relativity," *Philosophy of Science,* XXXIII, 4 (December 1966): 341–344.

† "Time and Physical Geometry," this JOURNAL, LXIV, 8 (April 27, 1967): 240–247.

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volving the auxiliary constructs *explicitly*. Thus Minkowski pointed out that what had been called "relativity of length," "relativity of simultaneity," etc., are best understood in the context of an "absolute" geometrical framework—but one within which not such statements as "a and b are length λ apart," for example, but such as "a and b are length λ apart in respect of T as time axis," have a well-defined meaning. And to signalize this shift in emphasis or context, Minkowski proposed that Einstein's term 'principle of relativity' be replaced by the term 'postulate of the absolute world'.

Minkowski himself still described his "absolute world" in a language dominated by the notion of *coordinates*—in the spirit of Felix Klein's famous "Erlanger program," which proposed that the intrinsic structure of a geometrical object be defined by specifying the group of coordinate transformations under which the formulas expressing that structure are to be invariant (or the quantities that enter the formulas are to be "covariant"). Since the coordinates are not themselves "intrinsic," this way of speaking is in effect circumlocutory; and therefore, although conceptually impeccable, it is deficient in lucidity. Moreover, there are contexts (of which, in physics, general relativity is one) where the Erlanger program, literally taken, does not work. Geometry, therefore, in the present century, has developed new techniques of formulation, which help both to achieve greater directness and lucidity of exposition and to deal with problems more simply than by the older techniques. Setting aside the latter point, the former is germane to our present concern. The following is an intrinsic characterization of Einstein-Minkowski space-time:

> Space-time is a four-dimensional real affine manifold, given together with a class of nondegenerate inner products on the associated vector space; this class is closed under multiplication by nonzero real numbers, and is generated (under this condition) by any one of its members; the members of the given class have index of inertia 1 or 3.

While it would not be appropriate to set forth in detail here the meaning of the terms used in this characterization, some indications are in order. That space-time is a four-dimensional real affine manifold may be taken to mean that it has a set of distinguished subclasses, called "straight lines," having, with respect to the incidence of points —hence also with respect to intersection and parallelism—, all the properties of the straight lines of a four-dimensional Euclidean space. A vector associated with this manifold may be thought of as a class of ordered pairs of points: the pairs (a,b) and (c,d) belong to the same

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class if and only if either $a = b$ and $c = d$ or, for some e and f, the quadrilaterals $abef$ and $cdef$ are parallelograms; addition of such vectors, and their multiplication by (first rational and then) real numbers, can be defined in a familiar way,[1] and the set of all vectors is then in the standard algebraic sense a four-dimensional real vector space. An inner product is a real-valued symmetric function of pairs of vectors, linear in each member of the pair; if we call two vectors orthogonal when their inner product is zero, nondegeneracy of the inner product means that the zero vector is the only one orthogonal to all vectors; while our condition on the index of inertia means that, for any four linearly independent and mutually orthogonal vectors, the inner products with themselves of three of these vectors have one sign (positive or negative), and the inner product of the fourth with itself has the opposite sign.

The arguments we shall have to examine deal with the notion of *time;* we must now consider how and to what extent this notion can be derived from the structure just specified. First, we call "timelike" any vector whose inner product with itself (using a product in the given class) is opposite in sign to that of any vector orthogonal to it; we call "null" any vector that is orthogonal to itself; and we call "spacelike" any vector that is neither timelike nor null. These notions are clearly intrinsic; they do not depend upon the choice of an inner product within the given class. Thus for any point a of space-time, all space-time points b are divided—in relation to a—into three mutually exclusive classes, according as the vector defined by (a,b) is timelike, null, or spacelike; and we say that b has timelike, null, or spacelike relation to a. In the natural topology of space-time (as a real affine manifold), the class of points in timelike relation to a is disconnected: it has two components, the two open convex lobes of the "light-cone" at a.[2] If a and b are any space-time points, and if A and B are, respec-

[1] In this order of development one wants to extend the multiplication from rational to real numbers "by continuity." This presupposes the construction, on the set of vectors, of a topology and a uniform structure for which both the addition of vectors and their multiplication by rational numbers are uniformly continuous. The required construction—on the basis of the assumptions about straight lines—is straightforward, but somewhat elaborate; and in a systematic mathematical treatment it is technically simpler to develop the theory in a slightly different way (starting from another characterization of "affine structure"). But the sketch given above *can* be carried through in detail, and the sketch should suffice both to indicate what the "intrinsic" theory looks like and to ground the concepts needed later. (As for the mathematical details, one could hardly improve upon the beautiful exposition in Weyl's book *Raum Zeit Materie.* It is unfortunate that this great work is available in English only in a translation garbled to the point of unintelligibility.)

[2] The light-cone is the set of all points in null relation to a; it is a cone with vertex at a, and separates space-time into three components: the set of points in

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tively, open convex lobes of the light-cones at a and at b, then we say that A and B *lie on corresponding sides* of a and b if the intersection of the sets A and B contains a complete open convex lobe of some light-cone. This notion, as a relation between A and B, is reflexive, symmetric, and transitive, i.e., an equivalence relation; and there are just two equivalence classes, each having as member one of the two open convex lobes of the light-cone at any point. The principles of relativistic space-time do not provide any way to single out one of these two classes, and the fundamental principle of such singling out remains to some extent a mystery from the point of view of physics as a whole; but in the empirical application of this space-time theory the two classes must be distinguished absolutely, for one represents the past and the other the future. To be precise: using the labels 'P' and 'F' for the two equivalence classes, we define the *past* of a, for any space-time point a, as the topological closure, in space-time with a itself removed, of that open convex lobe of the light-cone at a which belongs to P; and similarly we define the *future* of a to be the closure in space-time with a removed of the lobe belonging to F.[3] It is easy to show that a is in the past of b if and only if b is in the future of a, and therefore that "being in the past of" is an asymmetric relation; it is also easy to show that this relation is transitive, and so defines a partial ordering of space-time—we call this the "chronological ordering."

It is a very crucial fact in the Einstein-Minkowski theory that the chronological ordering of space-time is not a total ordering, even in the wider sense. By 'total ordering in the wider sense' I mean the following: If R is an asymmetric and transitive relation on a set S, the ordering it defines on that set is total in the narrow sense if, for any a and b in S, we have aRb or bRa or $a = b$; i.e., if the relation

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spacelike relation to a, and the two components of the set of points in timelike relation to a. Its name comes from the fact that it represents in a certain sense the locus of all possible direct electromagnetic influences ("light signals") toward or from a.

[3] To avoid misunderstanding, it should be emphasized that: (1) The structure of space-time as defined above affords no basis for singling out one class as P and the other as F: in so far as this distinction has to be made, one must regard it as an addendum to the former definition—i.e., as postulating a new element of structure. (2) The theory of relativity *per se*, and a large part (possibly all) of theoretical physics, can dispense with this additional structure: their principles can be formulated in such a way that the two equivalence classes enter symmetrically, with nothing to show which class is P and which is F. (3) The need in empirical application to distinguish P from F causes no difficulty, since this distinction is compatible with (although not determined by) the structure we have defined; but the contrast between the deep role of the distinction in experience and its apparent irrelevance to the basic principles of the physical world is very puzzling, and constitutes one of the major philosophical problems about time.

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"neither aRb nor bRa" is the same as *identity* (of a and b). Under the weaker condition that this relation is *an equivalence,* I call the ordering total in the wider sense. When this condition holds, we may always pass to the quotient set $\bar{S}$ under this equivalence relation (i.e., to the set of equivalence classes); it is easy to see that R induces on $\bar{S}$ an ordering $\bar{R}$ (such that if a and b in S belong respectively to the equivalence classes $\bar{a}$ and $\bar{b}$ in $\bar{S}$, then $\overline{aRb}$ if and only if aRb), and that this ordering of $\bar{S}$ is total in the narrow sense. Now, just this situation characterizes the pre-relativistic theory of time. According to that theory, space-time has a chronological ordering that is total in the wider sense; and the quotient set of space-time to which this gives rise—and on which the chronological ordering is total in the narrow sense—*is* just what is called "time." But no such natural decomposition of space-time occurs in the relativistic theory: although *temporal relations* have a meaning in this theory, *time* as a separate concept has no role and no meaning for it—so that Minkowski was led to say: "Henceforth space by itself, and time by itself, are doomed to fade into mere shadows, and only a kind of union of the two will preserve an independent reality."

Let us compare the relativistic and pre-relativistic space-time theories more systematically. Both theories assume that space-time has a four-dimensional affine structure. In the chronological decomposition of pre-relativistic space-time, the equivalence classes ("instants") form a family of parallel three-dimensional affine subspaces; and a straight line transverse to these (a "timelike" line) is chronologically ordered in agreement with its ordering as an affine (or topological) line. It follows that we can introduce an affine coordinate function that is constant on the equivalence classes, increasing from class to class in the direction from past to future: this is "the time coordinate," unique up to a change of scale and of zero point. Furthermore, according to the pre-relativistic theory, the three-dimensional space of each instant possesses a Euclidean metrical structure, the straight lines of which coincide with the straight lines of space-time (as an affine manifold) that fall within that instant. The significance of the affine structure—in both theories, relativistic and pre-relativistic—is that timelike straight lines of space-time represent possible uniform rectilinear motions, and parallel timelike straight lines represent motions with the same velocity.[4]

Now consider the following questions:

[4] In both theories it is possible to characterize nontimelike (i.e., spacelike or—in the relativistic case—null) straight lines in terms of timelike ones. (The underlying reason is that in both cases the vector space associated with space-time is *spanned* by timelike vectors.)

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(1) What is the time interval between space-time points a and b?
(2) What is the spatial distance between space-time points a and b?
(3) What is the spatial separation of two parallel timelike lines—for instance, the lines that represent the motions of two marked points on a measuring rod in uniform rectilinear irrotational motion (in other words, what is the *length* of the rod between those two points)?

In the pre-relativistic theory, question (1) has a definite answer always (assuming that a unit of time—i.e., a scale for the time-coordinate function—has been chosen). Question (2), in contrast, has a somewhat singular status. If the answer to (1) is *zero*—if, in other words, a and b are in the same instant—then (subject to the choice of a unit of length) question (2) has a definite answer, from the Euclidean structure of the instants. But if a and b do not belong to the same instant, then if we adopt the classical point of view known as "Galilean relativity," question (2) does *not* have a definite answer, and it is always possible to make the answer zero—that is, to declare a and b to be *the same point of space at different times*—by choosing the direction of the timelike vector defined by (a,b) as "time axis" in space-time (which is to say, by regarding the particular state of uniform rectilinear motion represented by this direction as the state of "rest"); for Galilean relativity asserts, within classical ("pre-relativistic"!) space-time theory, that there is no physical criterion to distinguish rest among all states of uniform rectilinear motion. Once a time axis has been chosen, the answer to (2) is determined; and varying the choice of time axis allows us to vary the answer *arbitrarily*. Finally, question (3) has a definite answer (independent of the choice of a time axis); for two parallel timelike lines intersect each instant in a pair of points for which the time interval is zero, hence for which the spatial distance is defined; and, on account of the parallelism, this distance remains constant. Thus, in summary, we have for the pre-relativistic theory: *questions* (1) *and* (3) *have definite answers; the answer to* (2) *is definite only when the answer to* (1) *is zero—otherwise it depends upon the choice of a time axis, and by suitable choice of an axis can be made to be anything we please.*

In the theory of Einstein and Minkowski, *none* of these questions, by itself, *ever* has a definite answer, except for the trivial case when $a = b$ or, in (3), when the lines coincide; otherwise, the answer to each is determined only upon the choice of a time axis.[5] If, indeed, we choose any timelike direction T, there is singled out, at

[5] The first part of Einstein's paper "Zur Elektrodynamik bewegter Körper" (1905) is devoted to establishing this dependence, on the basis of physical principles and of an analysis of the process of spatiotemporal measurement.

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each space-time point a, one unique distinguished three-dimensional "spacelike" affine subspace of space-time: namely, the set of all points b for which the vector defined by (a,b) is orthogonal to T. The spacelike subspaces thus distinguished form, like the instants of pre-relativistic theory, a family of parallel subspaces of space-time, and allow us to introduce an affine time coordinate (constant on these subspaces), which is then unique up to a choice of zero point and scale. Moreover, the scale can be fixed *once for all* (independently of the chosen axis T) by selecting one out of the given class of inner products: thus a change in the choice of time axis, which introduces a quite new time coordinate, does not require a new choice of scale.[6] If (as we may assume) the chosen inner product has index of inertia 1, its restriction to the set of vectors parallel to our spacelike "slices" will be a *positive definite* inner product, i.e., a Euclidean metric on those slices (our universal choice of time scale giving thus a spatial scale as well). In short, after choosing T we have—relative to T—on Einstein-Minkowski space-time all the structure we had in pre-relativistic space-time (with a time axis); therefore questions (1), (2), and (3) now have answers. When we examine the effect of varying T, we find the general situation to be this:

(1′) When b has timelike or null relation to a (and is not a):
if b is in the future of a, the answer to (1) can be made to be any strictly positive number;

[6] The theory of the "Lorentz time-dilatation" depends upon this circumstance. If b is in the future of a, selection as time axis of the direction defined by (a,b) minimizes the time-coordinate difference from a to b; change to another axis "dilates" the time interval.

The physical significance of this relationship deserves some further comment. It is strongly dependent on the notion of "proper time," which plays a central role in relativistic physics. Proper time is not a quantity attached to space-time points or to pairs of space-time points; it is in this respect a notion utterly different from the quantity "time" or "time interval" of pre-relativistic theory. Instead, proper time is associated with each timelike (straight or curved, but sufficiently "smooth") *line segment* in space-time. As attached to such line segments, it is of the nature of a *length,* defined "infinitesimally" by the square root of the absolute value of the inner product with itself of the tangent-vector to the line; and choice of a particular inner product amounts to choice of a unit or scale for proper time. The fundamental physical role of proper time comes from the principle (here stated roughly) that whenever a process takes place along a well-defined line of space-time ("world-line"), the time rates in the dynamical principles that govern that process are to be understood in terms of the proper time along that line (and *not* in terms of a "time coordinate"—except mediately, that is after the proper time has been expressed in terms of the coordinates). According to this principle, for example, a clock beats proper time. Since a suitably scaled time coordinate coincides with proper time for straight lines parallel to the time axis of that coordinate, it follows from the Lorentz dilatation as formulated above that choice of an axis for which the clock is in motion will make the clock (in comparison with the time coordinate) appear slow.

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if b is in the past of a, the answer to (1) can be made to be any strictly negative number.

When b has spacelike relation to a, the answer to (1) can be made to be anything we please (including zero).

(2′) When b has spacelike or null relation to a (and is not a), the answer to (2) can be made to be anything we please except zero;

when b has timelike relation to a, the answer to (2) can be made to be anything we please (including zero).

(3′) When the lines are distinct, the answer to question (3) can be made to be anything we please except zero.

It will be remarked that in all this nothing has been said about "observers": such a notion is not required for stating the theoretical relationships. But Einstein's analysis, which *established* these relationships, turns upon the fact that our procedures of spatiotemporal measurement single out—*de facto*—a particular state of motion: that of ourselves and our measuring instruments; and thus single out a time axis. It is in this sense that the quantities we measure are, intrinsically, *relative to a time axis;* and it is because the time axis in question is determined by "the observer" 's state of motion, that one often characterizes such notions as simultaneity, time interval, and length, in Einstein's theory, as "relative to the observer." As we shall see,[7] this way of speaking can lead to serious confusion. One should never lose sight of the fact that any observer, regardless of his state of motion, can express the results of his own measurements (and those of colleagues) in *absolute, physically meaningful language,* which is *the same for all observers.*

II

The geometrical fact underlying the arguments of the two papers under discussion is this:

> For any points a and b of Einstein-Minkowski space-time, there is a point c in spacelike relation to both: i.e., such that c is comparable neither with a nor with b under the chronological ordering.

Since for any pair of chronologically incomparable points there is a time axis assigning to the time interval from the one to the other any value we please, positive, negative, or zero [cf. (1′) above], it follows that: for some time axis, the values of the time-coordinate function at a and at c are equal; for some time axis, the values of the time-coordinate function at b and at c are equal; and for some time axis, the value of the time coordinate at b is smaller than the value at c.

From these facts Rietdijk argues as follows: We may conceive an

[7] Compare fn 15 below.

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observer A (with constant velocity), passing a, for whom a and c are simultaneous. We may conceive an observer C, passing c, for whom b is simultaneous with (or earlier than) c. Then for A at a, c is "now"; but for C at c, b is "now" (or even "already past"); therefore, the event at b is fully determined at the moment when A is at a.—Or again: we may suppose that C near c very rapidly changes his velocity (i.e., his distinguished time axis) so that it comes to agree for a moment with that of A, and that he then as rapidly changes it back. In the intermediate phase, C at c is at rest with respect to A, "experiencing the same 'present' " with him—a present to which both c and a belong. But *before* this, b was already present (or even past) for C. Therefore (according to Rietdijk) C can prove that nothing done by A at a could have had any kind of influence upon an event at b—and this may be taken as the rigorous meaning of the assertion that "the event at b is already determined at a." [8]—But a and b were any two space-time points; in particular, b may belong to the future of a (and —for emphasis—to the arbitrarily remote future). Rietdijk concludes that every event "is pre-determined 'from time immemorial' "; that determinism (which is not the same thing as causality) must therefore hold in microphysics; and that free will is impossible, and all philosophies that depend on free will are untenable.

Putnam's argument is different in detail, but essentially the same. It is based upon the premise, taken as part of "the 'man on the street's' view of the nature of time," that *all things that exist now are real,* and the premise that the relation "event x is real when event y occurs" (as a relation of x to y) is transitive; and it concludes, starting from a as "now," that since for some observer a and c are simultaneous, whereas for some observer c and b are simultaneous, the event at b must be real "now": that is (a and b being arbitrary), that everything (the entire history of the world) is always real.[9] In the same way, commenting upon Aristotle's example of "the sea-fight tomorrow," [10] Putnam concludes that propositions about the future have a determinate truth value in the present. And he suggests that these results

[8] As we shall see, Rietdijk is wrong: C cannot prove any such thing. But, further, the proposed explication of 'b is determined at a' is specious: if, for instance, we consider the possibility of a stochastic world—or an utterly capricious world—in which all events are entirely independent, we have on the one hand an extreme case of "inability to exercise an influence," and on the other hand what may very reasonably be considered an extreme case of indeterminism.

[9] This is the kernel of Putnam's discussion (p. 246); earlier in the paper he offers more discursive considerations (based upon the same geometry), leading to the weaker conclusion that *some* future events are real.

[10] In a minor scholarly lapse, Putnam gives Aristotle's example as "Who will win the sea-fight tomorrow?"; in fact Aristotle discusses whether there will be a sea-fight.

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constitute the decisive solution of the main philosophical problems about time.

The central conclusion of both papers, then, is that special relativity is incompatible with a notion of temporal evolution as (in some sense) *a becoming real, or becoming determinate, of what is not yet real or determinate.* But the arguments are incorrect, and the conclusion is mistaken. The notion of "becoming" may be epitomized as follows:

> For an event—a man considering, for example—at a space-time point *a*, those events, and only those, *have already become* (real or determinate), which occur at points in the topological closure of the past of *a*.[11]

This characterization can be taken over word for word from pre-relativistic to Einstein-Minkowski space-time; because the relation "being in the past of" is an ordering of space-time, the obvious compatibility requirement—that if *a* has already become for *b*, and *b* for *c*, then so has *a* for *c*—is satisfied. It is more or less apparent to simple inspection that the arguments of Rietdijk and Putnam summarized above contain nothing that can impugn this notion. I shall try presently to make explicit the fallacy of those arguments; but it will be useful, first, to examine an express attempt of Putnam's to dispose of this view of becoming.

Putnam addresses himself to this view in the form of a proposal that "only statements about events in the lower half of my light-cone have a truth value; only events that are in 'my past' according to *all* observers are determined." He offers two objections. The first is:

> This . . . move . . . flagrantly violates the idea that there are no Privileged Observers. Why should a statement's having or not having a truth value depend upon the relation of the events referred to in the statement to just one special human being, *me?* (246)

The answer is that "having or not having a truth value," in this question, must be understood classically to mean "at a given time" (the puzzle about the sea-fight tomorrow is whether there is a definite truth value *today*); but "at a given time" is not a relativistically invariant notion, and the question of definiteness of truth value, *to make sense at all* for Einstein-Minkowski space-time, has to be interpreted as meaning "definiteness at a given space-time point (or

[11] "Topological closure" because it is natural to say that something *in or on the boundary of* the past of *a* has "by now" become for *a*; note that *a* itself is thereby included (and, for Einstein-Minkowski space-time, besides the past of *a only a* itself). It hardly needs to be added that we do here definitely assume the distinction of past from future to be absolute (cf. fn 3 above).

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event)"—to be vivid: "definiteness for me now." The "Privileged Observer" (or, rather, privileged event) is—in effect—named in the question, and therefore has every right to be considered germane to the answer. Putnam's objection has an exact analogue, whose inappropriateness is plain, in the pre-relativistic case; namely, the question: "Why should a statement's having or not having a truth value depend upon the relation of the events referred to in the statement to just one special time, *now*?"—Putnam bases his second objection upon the remark that an object—a man named Oscar, for instance—can have a world-line entirely in spacelike relation to "me-now," but intersecting the past of "me-later." This is true (and it is even possible for all the points of "Oscar" 's world-line to have spacelike relation to *all* the points of "my" world-line).[12] Putnam derives, for the notion of becoming, this "highly undesirable consequence":

> ... when that future becomes the present, it will become true that Oscar *existed,* although it will never have had a truth value to say in the present tense "Oscar exists now." [13] Things could come to *have been,* without its ever having been true that they *are!* (246)

But his conclusion is by no means as startling as Putnam seems to think; to reach that conclusion, at any rate, we do not at all need his ingenious example. If "presentness to each other" of events is taken to mean that *each has, for the other, already become,* then in pre-relativistic theory we have the ordinary concept; but in Einstein-Minkowski space-time *an event's present is constituted by itself alone.* In this theory, therefore, the present tense can *never* be applied correctly to "foreign" objects. This is at bottom a consequence (and a fairly obvious one) of our adopting relativistically invariant language —since, as we know, there is no relativistically invariant notion of simultaneity. The appearance of paradox only confirms that the space-time of Einstein and Minkowski is quite different from pre-relativistic space-time.[14]

Let us turn now to the main arguments of Rietdijk and Putnam. The common fallacy of those arguments is their employment, in the

[12] For objects of finite duration and spatially remote, this can happen easily; for objects of unlimited duration, it can happen only in rather bizarre physical circumstances (involving relative velocities asymptotically approaching that of light).

[13] *Lapsus calami* for "it will never have been true to say ..."? This seems to be the only reasonable interpretation.

[14] The paradox is mitigated when we observe that, for processes of more than instantaneous duration, a meaningful and intuitively satisfying notion of "contemporaneity" can be defined: two such processes may be said to be contemporaneous if part of each is past to part of the other—in other words, if mutual influence ("communication") is possible between them.

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context of the Einstein-Minkowski theory, of notions about time that are illegitimate in that theory. As we have seen in sec. 1, the Einstein-Minkowski structure gives us (in a very clear mathematical sense) *temporal relations,* but no "time" *simpliciter.* In the context of special relativity, therefore, we cannot think of temporal evolution as the development of the world *in time,* but have to consider instead (as above) the more complicated structure constituted by, so to speak, the "chronological perspective" of each space-time point. The leading principle that connects this mathematical structure with the notions of "process" and "evolution" (and justifies the use of our notion of "becoming" in relativistic space-time) is this:

> At a space-time point a there can be cognizance of—or information or influence propagated from—only such events as occur at points in the past of a.

In the geometrical situation defined at the beginning of this section, the point c is not in the past of a (nor identical with a), and a is not in the past of c; therefore, no information or influence from or cognizance of either can occur at the other. This provides no warrant at all for the claim that the event at c must be "real" or "determinate" to an observer at a; and the fact that there is a time axis orthogonal to the direction from a to c (or a time-coordinate function having equal values at a and c) adds nothing. The relation of b to c is of course just the same; the existence, adduced by Rietdijk, of a time coordinate whose value is smaller at b than at c, does not put b—in the absolute, or physical, or relativistically invariant sense—in the *past* of c, because *"a time coordinate" is not "time."* [15] Neither a nor b is, in any physically significant sense, "present" (or past) for any observer at c—regardless of his velocity—for neither *has already become* for c (nor has c for them); but a has already become for b, and can influence it.[16]

[15] The failure to recognize this very basic distinction probably comes from a wrong understanding of the description of temporal concepts as "relative to the observer." Both writers try to make "the observer" carry with him, as it were, a pre-relativistic temporal structure, with "his" time coordinate playing the role of time. But what Einstein's arguments showed was that *a certain procedure of measurement singles out a time axis and gives numerical time differences dependent upon that distinguished axis;* not that an observer's state of motion imposes upon him a special view of the world's structure. This illegitimate metaphysical interpretation of the time coordinate appears perhaps most plainly in Rietdijk's phrase describing C and A, when at rest with respect to one another, as "experiencing the same 'present' "; there is of course no such "experience": the fact that there is no experience of the presentness of remote events was one of Einstein's basic starting points.

[16] In saying this, of course we set aside all philosophical reservations about the notions of "cause" or "influence" that do not depend upon the theory of relativity.

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III

Rietdijk's paper is straightforward: it proposes an explication of the notion "predetermined" (which, however, seems to me to be faulty); [17] and it proceeds to demonstrate (fallaciously, as I believe I have shown) that special relativity conjoined with this criterion implies determinism. But the philosophical framework of Putnam's paper is more complex, and I should like now to consider certain issues of philosophical method that it seems to me to raise.

We have seen that one of Putnam's principal starting points is what he calls "the 'man on the street's' view of the nature of time" (240). This view—or rather, as he says, "the main principle underlying his [the man on the street's] convictions on this subject"—he expresses as follows: "All (and only) things that exist *now* are real." Two things about this starting point are striking. First, Putnam's argument makes crucial use of the special theory of relativity, which itself *began* from a *critique* of the ordinary notion of time; it therefore seems strange to appeal in this argument to the "man on the street" as an expert on time. Second, Putnam's conclusion—that all things, for all time, are real—contradicts the "man on the street's view." What sort of argument is it, then, that contradicts its own premise?

From the strictly deductive point of view, such an argument is a *reductio ad absurdum,* i.e., a refutation of the conjunction of its premises; and if we consider the "man on the street's view" to be the only questionable premise of the argument, the conclusion is that the man on the street is in error. I think this is the correct conclusion; and in light of the first of the preceding two remarks it is not very surprising. But an argument whose conclusion is incompatible with its premises may be viewed in another way: as a heuristic, rather than a strictly deductive, argument. The major difference is that whereas the formal "conclusion" of a *reductio ad absurdum* is of no interest in itself (as being "absurd"), a heuristic argument can lead to the acceptance of a positive conclusion which, in some sense, *corrects* its premises. The principles of such argumentation are notoriously hard to codify; and it seems to me that, just for this reason, if one regards the process as a serious one—and not just a frivolous exercise or game—, one is obliged in pursuing it to seek the greatest possible clarity at all stages.

Now, Putnam's argument must evidently be regarded as of this second kind: he cannot merely intend to refute the view of the "man on the street" (or even of Aristotle); for he says, "I conclude that the problem of the reality and the determinateness of future events is

[17] See fn 8 above.

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now solved" (247). What, then, is the logic of this heuristic or dialectical argument?—In fact, having stated the "man on the street's view" that "real" means "presently existing" and having correctly shown that this is incompatible with special relativity (if one assumes 'real' to have an objective meaning—i.e., to be relativistically invariant), Putnam bases the rest of his discussion upon "our desire to preserve . . . one-half" (243) of that view: namely (cf. sec. II above), the principle that all presently existing things are real. But why do "we" have this desire, and how legitimate is it? The first starting point —the view of the "man on the street"—is prima facie questionable. A dialectical argument with such a starting point may lead to results that are both interesting and convincing; but, clearly, the convincingness has to come from something besides the dubious starting point itself. Putnam, however, offers us no authority higher than the "man on the street"; on the contrary, having first refuted the view of the latter (242), he quite casually proposes (243) to save one-half that view. It is amusing to ask, Why not try to save the other half? If, indeed, instead of maintaining the implication "present implies real," we were to insist upon the converse—i.e., that *only* things that exist now are real—, we should be led by an argument like Putnam's to conclude that *for any event, it and it alone is real;* and then, instead of the interesting result that special relativity implies determinism, we should have the interesting result that special relativity implies a peculiarly extreme (but pluralistic!) form of solipsism.

IV

Putnam's argument uses (apart from the geometry of relativistic space-time) a second premise: namely, the requirement that "x is real for y" be a transitive relation. This—as a leading principle, and part of an explication of the idea of "reality"—is very reasonable. But Putnam bases it upon the following assumption, which he calls "the principle that There Are No Privileged Observers" (thus associating it in a sense with relativity theory itself):

> If it is the case that all and only the things that stand in a certain relation R to me-now are real, and you-now are also real, then it is also the case that all and only the things that stand in the relation R to you-now are real (241).

This, however, must not be interpreted literally; for taking R to be such that xRy if and only if x is real and y is me-now shows that the principle as stated is unreasonable. Putnam actually qualifies it in two important ways: "R must be restricted to physical relations that are supposed to be independent of the choice of a coordinate system . . . and to be definable in a 'tenseless' way in terms of the funda-

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mental notions of physics. And . . . it must not depend on anything *accidental* . . . that all and only the things that stand in the relation R to me-now are real" (241).

I find this formulation quite obscure. It is particularly puzzling that, after the relation R has been explicitly required to be "physical," "independent of the choice of a coordinate system," and independent of anything "accidental," there should remain any need for a *further* requirement of "No Privileged Observers"; why is this not sufficiently guaranteed by the relativistically invariant character of R and its independence of accidental particulars? There is also something disturbing in the statement of the principle: why is the notion "real" assumed, on the one hand, to be characterized by a relation R, whose referent is clearly something like an event or space-time point (or an observer at a given time); while it itself, on the other hand, is treated as a simple predicate, unqualified by any such reference?

If I were to reformulate the principles that Putnam seems really to mean in this passage, I should do so as follows: First, the notion "real," for the arguments of the paper, must be represented by a relation ("x is real for y"), either between events or between space-time points; for simplicity, let us say the latter; and let us call this relation "R." Then we assume that R is definable, within space-time theory, in intrinsic geometrical terms; this ensures its independence of coordinate systems, of privileged observers, and of anything "accidental"; and *in addition* we assume (guided by our primitive notion of "reality") that R is transitive.

This, however, although it seems to satisfy all the criteria employed by Putnam to justify his assumption, does not imply that assumption itself. Our principle says, "If yRz and xRy, then xRz"; his says (taking z to be "me-now" and y "you-now"), "If yRz, then xRy if and only if xRz." Since Putnam also assumes—and we ought to follow him, adding a further assumption to the foregoing—that R is reflexive, one easily sees that in effect what he calls the principle of No Privileged Observers just requires R to be *an equivalence relation.* But such a requirement has in fact no connection with the privilegedness of observers; and it is moreover entirely inappropriate to Einstein-Minkowski space-time—in which (unlike pre-relativistic space-time, with its temporal decomposition) *there are no intrinsic geometrical partitions into equivalence classes at all, besides the two trivial ones:* that into just one class (all of space-time), and that into classes each consisting of a single point. (This theorem curiously echoes our previous alternative: "determinism" or "solipsism.")

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All this has no essential effect upon Putnam's main argument, in which he uses only the *transitivity* of R. But when he takes up that view of becoming which is in fact compatible with special relativity, then, as we have seen, he dismisses it as a flagrant violation of "the idea that there are no Privileged Observers." I have shown that there is no such violation. There is, however, a violation of what Putnam has introduced, quite illegitimately, under that name; for the relation "being in the past of or identical with"—which plays the role of R in this view—is intrinsic, transitive, and reflexive, but is (of course!) not an equivalence relation.

V

Putnam thus has used, to interpret a formally correct geometrical argument, a collection of principles of philosophical interpretation derived from dubious authority, subjected to no adequate critical examination, and in point of fact demonstrably inappropriate to the scientific context in which they are employed. Putnam's good work prevents any suspicion that these defects are characteristic of him: they seem to me to merit comment precisely as symptoms of a *prevalent* laxness in philosophical discussion, which is very much to be deprecated. Technical mistakes and errors of detail can be corrected; differences in principle, when they are examined in an objective spirit and brought to a condition of clarity, can be fruitful, and illuminating; but the lowering of critical standards in philosophical discourse itself precludes understanding and is the death of philosophy.

How far this is true in the present case appears, I think, most clearly when we turn from the principles and methods of argumentation to the conclusions themselves. In presenting, above, the relativistic notion of "becoming," I have not intended to argue in favor of "indeterminism" as opposed to "determinism," but only to show that the papers under discussion cannot be considered to settle the issue. But the issue itself is singularly obscure, and, despite their decisive claims, the two papers make not the slightest attempt to clarify its *meaning* (except, perhaps, for Rietdijk's proposed criterion, which may be regarded as a partial attempt). Historically—certainly, for instance, in Aristotle—the issue of "determinism" has been associated with what in modern times is called the notion of *causality:* that is, determination *of the future by the past.* Now, even if we do consider the entire history of the world to be (in some sense) always "real" or always "determinate," it in no way follows that *knowledge of some part* of that history—namely, the past—suffices to provide *knowledge of the rest.* And neither Rietdijk nor Putnam draws this

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conclusion (Rietdijk, as we have seen, disclaiming it expressly). Therefore, if it is reasonable to suppose that our direct information is always, fundamentally, information about the past (or present), these authors require us to believe in the "determinateness" of the answers to questions when—not merely in practice, but in principle —information to decide those questions is inaccessible to us. Nor is this just a remote "philosophical" possibility: Rietdijk and Putnam are committed to hold that, for instance, the point of absorption of a single diffracted photon—which, on the current version of quantum mechanics, is *strictly unpredictable*—has been "determined" from all time.

Now, it is seriously questionable whether the view that *the future is determined, but (in principle) unpredictable* is different, as a view about the world (and not merely as a manner of speaking), from the view that events are indeterminate until they have occurred. Introduction of the logical notion of "truth" for propositions about the future does nothing to sharpen the issue. The statement "A sea-fight will take place tomorrow" is true if and only if a sea-fight will take place tomorrow. If, tomorrow, a sea-fight does take place, we shall know that that sentence is and was true; for this is what we mean by its "truth"—whether or not we hold that the correctly predicable truth value is already determinate today. On this point, Putnam objects that "It is unfair to assume a form of language which presupposes that Aristotle was wrong and then to use the assumed correctness of this linguistic formalism as an argument against Aristotle" (244). I agree; but I do not think that the notion of truth as a "timeless" predicate does prejudice the case against Aristotle, because I do not see that adoption of this notion—as a manner of speaking—says anything about the world. It is true that a manner of speaking may lead to mistakes by *suggesting* something about the world not legitimately implied by it; we have seen an example, in the misinterpretation of the usual language of relativity. But, although language can be refined, the only fundamental guard against such mistakes is clarity *of thought*. That misinterpretation of relativity does not reflect upon Einstein, who provided us with a sufficiently clear analysis: namely, clear enough *to allow us, with care, to avoid misunderstanding;* after that the responsibility was ours. Einstein's analysis is a paradigm of the clarity for which philosophy should strive; and which, in the Rietdijk-Putnam thesis of determinism, it has certainly not attained.

Far, then, from deciding the main philosophical questions about time, these papers fail even to stake out a philosophical position with

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any internal clarity. My own view is that the issues of "reality" and "determinism" in the Rietdijk-Putnam form probably *cannot* be clarified—i.e., that determinism apart from causality is a pseudo-problem. But I believe that very real and serious problems about time do still exist: not only the problem of past and future [18] (which Putnam too mentions as one that remains), but also a collection of difficulties—not yet easy even to formulate—connected with quantum mechanics (and especially with relativistic quantum mechanics). I see no reason to expect such problems to be solved at one stroke; and I agree wholeheartedly with Putnam in looking to physics, rather than "philosophy," for decisive progress. But the clearest possible statement of the questions seems to me a present philosophical task of some moment (although, again, perhaps one that members of scientific departments are as likely to carry out as members of philosophy departments). I should like, as a kind of postscript, to indicate in a rough way one such question, which is largely independent of the theory of relativity.

We ordinarily suppose (cf. above) that our information, or evidence, is about the past. But, in a sharper sense, it is fairly clear that our truly current evidence, at any moment, depends only upon or is contained in the *present* state of the world. The information obtained, for instance, from yesterday's experiment is current information only in so far as it is now written down (or remembered—but current remembrance too is part of the present state). Of course relativity makes some difficulties about the notion of "present" here; but (in this rough sketch) let us disregard them. The question I wish to raise is based upon the remark that the definiteness of our knowledge of the past is limited, in principle, by the definiteness with which inference is possible, on the basis of the laws of nature, from the present to the past.[19] But such inference, when it is founded on the laws of quantum mechanics, is subject to the same limitation to *statistical* assertions as is quantum-mechanical inference to the future; and (what seems much more serious) the notion of external intervention (by the process of measurement), which plays an essential role in the now standard interpretation of quantum-mechanical assertions, is not available at all for inference to the past. The first point suggests

[18] See fn 3 above, and cf. fn 11.

[19] Of course we do not ordinarily "know the past" *by means of* such inferences, any more than we ordinarily know the visual properties of objects by inference from retinal images and the laws of optics. But, as the laws of optics determine what we can possibly discriminate through vision, the laws of nature in general—both those we know and those we do not know—must determine what we can possibly discriminate of the past through our present state.

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the notion of a progressively increasing *physical uncertainty* (or "indeterminism") of the past: a kind of fading of the world's memory. The second point, however, seems to mean that some sort of clarification in principle is required before quantum mechanics can be applied at all in solid reasoning upon such questions.

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BOOK REVIEWS

The Philosophy of Leibniz. NICHOLAS RESCHER. Englewood Cliffs, N.J.: Prentice-Hall, 1967. iv, 168 p. $2.95.

A number of obstacles make Leibniz's thought relatively inaccessible to the contemporary student. Among these is the factor to which Leibnizians point as the source of much of the philosopher's continuing fascination: Leibniz's mature system is forged out of new and old ideas from an extraordinary range of disciplines and traditions, and directed in sometimes unobvious ways at a very wide variety of philosophical problems. Add to this consideration the notorious difficulty of finding one's way around the heterogeneous, huge, multilingual Leibnizian corpus, and the potential value of good, general secondary works on Leibniz's philosophy becomes apparent. And so, of course, does the array of qualifications requisite to produce such a work.

Professor Rescher's stated purpose is to provide a "systematic introduction" to Leibniz's philosophy. As an eminent philosopher, with an expert knowledge of mathematical logic, the history of philosophy, and, especially, of Leibniz's writings, he is unusually well qualified for the job. Equally significant is the fact that he admires Leibniz and says so, forthrightly introducing the subject of his study as one of his "philosophical heroes." These qualifications are reflected in the book, which opens with an unpretentious and rather moving account of Leibniz's life, and manifests throughout a stimulating appreciation of both the determinants and the content of Leibniz's positions. Rescher points out some difficulties and ambiguities in Leibniz's views, but his approach is in general sympathetic and unpolemical. The book contains much valuable material, including numerous off-the-beaten-track references to Leibniz's writings, and a new presentation, in a more comprehensive context, of interpretations and arguments from the author's excellent article, "Contingence in the Philosophy of Leibniz." [1] (Rescher's exposition of Leibniz's doctrine

[1] *Philosophical Review,* LXI, 1 (January 1952): 26–39.

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